

This artist's impression shows an unusual binary star system known as J0806, 1,600 light-years away from Earth. In this rare example, two dense white dwarf stars orbit one another.

ON MARCH 28, GERMAN ASTRONOMER WILHELM OLBERS (1758–1840) discovered the asteroid **Pallas** in an orbit similar to that of Ceres (see 1801). He incorrectly thought that Pallas, Ceres, and other asteroids discovered later were remnants of an exploded planet.



JEAN-BAPTISTE LAMARCK (1744–1829)

Originally a bank worker and army officer, Jean-Baptiste Lamarck became interested in botany and published the popular book *French Flora* in 1773. He also wrote a history and classification of invertebrates and became a highly regarded taxonomist. Lamarck is best known for his theory that acquired traits can be inherited by future generations.

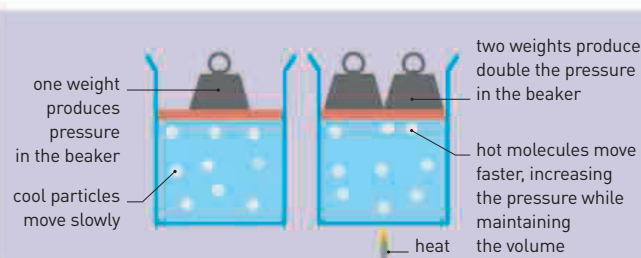
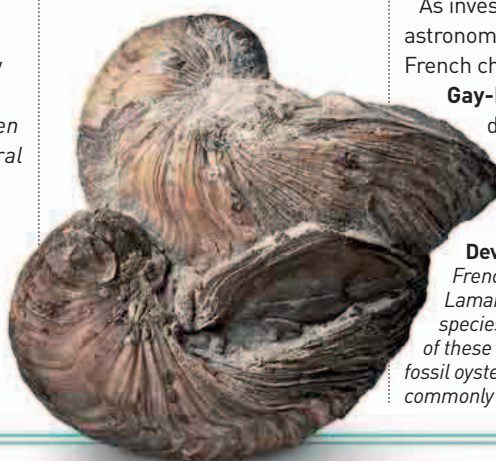
While carrying out experiments with glass prisms, **William Hyde Wollaston** noticed **dark lines in the spectrum of the Sun**. Although it was not known at the time, these lines indicate the absence of particular colors in sunlight. They are usually called **Fraunhofer lines**, after German physicist Joseph von Fraunhofer (1787–1826), who independently rediscovered the lines and studied them in more detail in 1814. In the 1860s, German scientists Gustave Robert Kirchhoff (1824–87) and Robert Bunsen (1811–99) showed how such lines functioned as fingerprints of different elements.

The study of evolution continued at a steady pace. **Jean-Baptiste Lamarck** became one of the first to use the word **biology** in its modern sense. German naturalist and botanist Gottfried Reinhold Treviranus (1776–1837) also used this term, independently of Lamarck, in *Biologie oder Philosophie der Natur Lebenden* (*Biology or Philosophy of Natural Living*). Both Lamarck and Treviranus came up with the idea that evolution occurs through the **inheritance of acquired characteristics**—a concept that came to be known as **Lamarckism**. This concept was often

demonstrated by citing the example of the giraffe: a giraffe that stretches its neck to reach high branches in the search for food will bear offspring that have a longer neck as a consequence (see 1809). Although incorrect, this concept was a step toward understanding evolution.

British inventor **Thomas Wedgwood** (1771–1805) was responsible for an important development in the field of photography. Using silver nitrate, he made the first **permanent images**—photographs in the form recognized today. British chemist Humphry Davy (1778–1829) described Wedgwood's work in the *Journal of the Royal Institution* in 1802.

The term **binary stars** was coined by German-born British astronomer **William Herschel** (1738–1822) in this year. A binary



THIRD GAS LAW

Described by French chemist Joseph Gay-Lussac, this law states that for a particular mass of gas at a constant volume, the pressure and temperature are proportional to each other. In the example shown here, when the pressure on the gas in the beaker is doubled, the temperature reached by the gas as it holds up the weights increases proportionately.

is a system where two stars—referred to as binary stars—orbit around each other. These types of stars are distinct from double stars that, despite being very far apart, lie almost along the same line of sight.

As investigations into astronomy gathered pace, French chemist **Joseph Gay-Lussac** (1778–1850) described a law that concerned the **behavior of gases**.

Devil's toenail

French biologist Jean-Baptiste Lamarck named numerous species in his lifetime. One of these was the Mesozoic fossil oyster *Gryphaea arcuata*, commonly known as Devil's Toenail.

According to this law, the pressure of a fixed amount of gas at a fixed volume is directly proportional to its temperature (on the scale that is now known as the Kelvin scale). At first, this law was named Gay-Lussac's Law, but because it follows directly from earlier work by French chemist Jacques Charles and Anglo-Irish chemist Robert Boyle, it is now called the **Third Gas Law** (see panel, above). Today, there is another, quite separate, law known as Gay-Lussac's Law.

Wollaston discovers dark lines in the solar spectrum

Lamarck and Gottfried Reinhold Treviranus promote the idea of inheritance of acquired characteristics

Humphry Davy describes Thomas Wedgwood's method of creating photographs using silver nitrate

April 10 The Great Trigonometric Survey of India begins with the measurement of a baseline near Madras

March 28 Wilhelm Olbers discovers the asteroid Pallas

June The first pediatric hospital opens in Paris, France, on the site of a former orphanage

British mathematician John Playfair publishes *Illustrations of the Huttonian Theory of the Earth*

William Herschel uses the term **binary star** for the first time to refer to a star that revolves around another star

Joseph Gay-Lussac demonstrates the **Third Gas Law**